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Semantic perception 3DGS-SLAM system for dynamic scenes

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Abstract

The integration of 3D Gaussian Splatting (3DGS) with visual simultaneous localization and mapping (SLAM) enables the construction of high-fidelity scene maps. However, most existing methods are designed for static environments, exhibiting significant performance degradation in dynamic scenes. To address this critical limitation, this paper proposes a novel semantic-aware SLAM system (GSV-SLAM) for dynamic scenes. Specifically, GSV-SLAM first deploys a YOLOv11-based semantic segmentation network to identify and remove predefined dynamic objects from input frames. Subsequently, it extracts ORB features exclusively from static background regions. This optimized processing sequence effectively reduces redundant feature computations while maintaining real-time performance. Second, an adaptive geometric consistency constraint strategy is employed to further suppress dynamic feature points on potential moving objects and reconstruct occluded background information caused by moving objects. Finally, a Gaussian map is constructed using optimized camera poses and static point clouds to achieve high-quality 3D reconstruction and new viewpoint rendering. Experimental results on the TUM and BONN datasets demonstrate that the proposed method achieves significantly lower absolute and relative trajectory errors compared to ORB-SLAM3. Real-world experiments further validate the system's robustness in scenarios with strong dynamic disturbances.

1. Introduction

Simultaneous Localization and Mapping (SLAM) technology serves as the core foundation for achieving autonomous navigation in mobile robots, positioning in autonomous driving systems, and environmental perception. Within the field of robotics, SLAM plays an indispensable role in environmental perception and autonomous positioning—whether for indoor inspections by ground-based mobile robots or stable locomotion by quadrupedal robots across complex terrains [1]. Among these, Visual Simultaneous Localization and Mapping (V-SLAM) [2], leveraging advantages such as low camera costs and rich visual information, provides critical technological support for autonomous perception and decision-making in intelligent equipment like mobile robots. It has gradually achieved widespread application across multiple domains including autonomous driving, drone navigation, and 3D reconstruction. Over the years, various visual SLAM systems have demonstrated remarkable performance under their respective applicable conditions. Representative achievements include systems such as ORB-SLAM3 [3], DSO [4], and VINS-Mono [5].

However, in real-world operating scenarios, moving objects frequently exist within the camera's field of view, which can easily cause visual SLAM systems to mis-track or lose feature points. This leads to cumulative positioning errors and may ultimately result in erroneous navigation decisions. To ensure accurate pose estimation, dynamic features must be effectively distinguished and filtered out from static features. In the field of visual SLAM, dynamic feature detection methods can be primarily categorized into two types: deep learning-based approaches [6] and methods combining geometric constraints with deep learning [7]. However,

approaches relying solely on deep learning typically demand substantial computational resources, making it challenging to meet real-time requirements. Methods integrating deep learning with geometric constraints have emerged as a promising research direction [8]. Simultaneously, mapping mechanisms operating under static scene assumptions struggle to handle dynamic objects effectively, often resulting in ghosting artifacts or erroneous geometric residues for moving targets in the map. 3D Gaussian reconstruction techniques [9] fundamentally overcome the limitations of traditional methods by parameterizing scene representation through differentiable 3D Gaussian distributions. Current research has made significant progress in dynamic scenes and 3D reconstruction, yet it still faces severe challenges. Most methods struggle to achieve an effective balance between positioning accuracy and real-time performance, often relying excessively on prior semantic information. Neural rendering methods, exemplified by 3D Gaussian reconstruction, have made notable advances in real-time rendering. However, their optimization process heavily depends on the quality of multi-view image inputs and still suffers from issues like geometric distortion in weakly textured regions. To address these issues, this paper proposes a Semantic-Enhanced Gaussian Splash Visual SLAM (GSV-SLAM) system aimed at achieving robust pose estimation and high-fidelity 3D reconstruction in dynamic scenes. The system constructs a semantic-aware real-time dynamic processing framework by deeply integrating sparse visual odometry with differentiable Gaussian splash mapping. At the semantic perception level, a lightweight modified YOLOv11 [10] network extracts dynamic object masks. The Slim Neck architecture balances accuracy and efficiency, enabling efficient removal of dynamic region feature points based on these masks. At the geometric level, a joint feature rejection algorithm integrating epipolar geometry constraints and depth consistency is proposed, effectively addressing the limitation of traditional methods in identifying non-prior semantic dynamic object feature points. Within the mapping thread, the ORB-SLAM3 framework is integrated with Gaussian splatting techniques to achieve real-time, high-precision 3D reconstruction under dynamic interference suppression. This system maintains real-time performance while significantly enhancing positioning robustness and reconstruction integrity in dynamic environments.

The core contributions of this paper can be summarized in the following three aspects:

- (1) **Lightweight semantic segmentation network:** An improved lightweight YOLOv11 instance segmentation model is proposed, which substantially lowers computational and parametric demands with minimal accuracy loss, allowing for efficient dynamic target recognition.
- (2) **Adaptive geometric consistency constraints:** A dynamic feature filtering mechanism is proposed, which integrates epipolar geometry constraints and depth consistency constraints. This complementary strategy detects and removes potential dynamic features points that fall outside the coverage of semantic masks.
- (3) **Semantically Guided Gaussian Mapping Strategy:** Combines multi-view geometric recovery with semantic smoothing filtering to complete occluded regions and suppress dynamic artifacts, generating structurally clear, sharply bounded, and complete 3D Gaussian maps.

Evaluation on the TUM RGB-D dataset and real-world moving scenarios shows that the proposed method surpasses current SLAM techniques in terms of both precision and robustness.

2. Related work

2.1. Deep learning-based methods

With the advancement of deep learning technology, object detection and semantic segmentation methods have been extensively integrated into dynamic visual SLAM systems to enhance their perception capabilities and robustness in complex environments. By fusing semantic information, these approaches effectively identify and exclude dynamic regions, thereby mitigating the adverse effects of moving objects on pose estimation and map construction. DLD SLAM [11] proposes an RGB-D SLAM method designed for indoor dynamic environments. By integrating semantic information with depth data, the system can identify and remove dynamic regions in real time, thereby enhancing localization accuracy and improving mapping stability in complex scenes. DP-SLAM [12] employs a Bayesian-based feature point motion model that estimates the dynamic probability of each point by combining distances and geometric constraints between latent dynamic features and mask boundary points, enabling more precise dynamic feature removal. RS-SLAM [13] generates dynamic masks by fusing semantic information and geometric constraints through probabilistic modeling. This approach effectively removes dynamic regions while preserving static feature points, thereby enhancing localization and mapping accuracy in complex dynamic environments. DIG-SLAM [14] combines YOLOv7 instance segmentation with motion consistency detection to eliminate dynamic point and line features from scenes. This reduces interference from moving objects on pose estimation and improves map consistency, but

its high computational complexity limits its application in high-frame-rate or resource-constrained scenarios. YDD-SLAM [15] employs YOLOv5 for object detection and depth analysis to mitigate interference from dynamic feature points. It integrates multiple optimization strategies to enhance visual SLAM stability in dynamic environments. However, its bounding box-based detection mechanism may lead to false removal of some keypoints.

To further enhance the effectiveness of dynamic feature point removal, researchers have conducted extensive explorations focused on semantic mask optimization. DYR-SLAM [16] combines RTAB-Map multi-frame point cloud information to optimize the YOLOv8 mask, while enhancing dynamic feature removal through acceleration constraints and multi-frame consistency detection. This approach improves pose estimation accuracy and dense point cloud map quality in complex dynamic scenes. YOD-SLAM [17] generates dynamic masks by combining YOLOv8 with depth information, then expands and partitions these mask regions to design differentiated feature rejection strategies for distinct areas, effectively mitigating dynamic interference on pose estimation. YLS-SLAM [18] introduces a boundary semantic enhancement mechanism based on semantic segmentation to progressively refine the edges of dynamic object masks, mitigating feature retention errors caused by imprecise segmentation.

Deep learning methods can effectively process recognized objects using prior information, but their performance relies on predefined categories, making it challenging to handle potential dynamic objects caused by human factors. In complex scenes, relying solely on semantic information remains insufficient for achieving stable and reliable dynamic feature filtering.

2.2. Methods based on deep learning and geometric constraints

In recent years, researchers have focused on integrating deep learning methods with geometric constraints to overcome the limitations of each approach, thereby achieving more precise dynamic feature point removal. Geometric approaches, such as reprojection error and polar line geometry, effectively identify and eliminate potential dynamic features, mitigating their interference with the system. Representative works include: DynaSLAM [19] employs Mask R-CNN [20] for pixel-level semantic segmentation to remove known dynamic objects, while integrating multi-view geometry to further eliminate potential dynamic points; SG-SLAM [21] targets dynamic scenes by constructing a joint feature mask integrating semantic information and geometric constraints, while designing a regionalized dynamic feature removal strategy to significantly reduce dynamic interference on camera pose estimation. OVD-SLAM [22] achieves accurate segmentation of foreground dynamic objects and background static regions by fusing optical flow, semantic information, and depth data, thereby enhancing visual SLAM robustness in dynamic scenes. DynPL-SVO [23] employs a dynamic grid algorithm to detect outliers, optimizing pose estimation through a joint cost function that integrates vertical and parallel reprojection errors from dynamic features. DSG-SLAM [24] proposes a visual SLAM framework combining dynamic feature rejection with dense reconstruction, removing dynamic feature points through motion consistency detection to reduce interference in pose estimation. SFD-SLAM [25] utilizes saliency analysis to identify potentially moving regions and combines geometric consistency to eliminate unreliable feature matches, thereby enhancing positioning stability in dynamic environments. However, its saliency detection may cause false positives in areas with varying illumination or complex textures, potentially compromising the accuracy of dynamic region identification.

2.3. 3DGS-based visual SLAM

In recent years, the 3D Gaussian Splatting (3DGS) technique, leveraging GPU parallel rendering capabilities, has rapidly emerged, offering new opportunities for real-time scene reconstruction and high-fidelity visual rendering. This method employs Gaussian distributions as core representation units, achieving differentiable rendering and ultra-high-frequency drawing by explicitly modeling illumination and geometric information in continuous space. It demonstrates remarkable potential in the field of visual SLAM. With the continuous advancement of 3DGS theory and engineering implementation, multiple SLAM systems based on its principles have been proposed. For instance, GSORB SLAM [26] utilizes ORB features combined with Gaussian splatting and transmission information for dense SLAM construction, achieving precise camera pose estimation and map reconstruction in highly detailed scenes. Gaussian SLAM [27] constructs a photo-realistic dense SLAM method based on Gaussian splatting, achieving high-quality scene reconstruction and visual localization by optimizing point cloud representation. GS SLAM [28] employs 3D Gaussian splatting technology to construct a dense visual SLAM framework, significantly enhancing the rendering quality and detail representation of dense point cloud maps while maintaining precise pose estimation. CG-SLAM [29] embeds dense RGB-D information into Gaussian fields within a unified uncertainty modeling framework, proposing a map optimization mechanism based on consistency estimation to achieve a balance between high accuracy and high efficiency. Gs3LAM [30] incorporates semantic information into Gaussian splatting modeling, constructing a

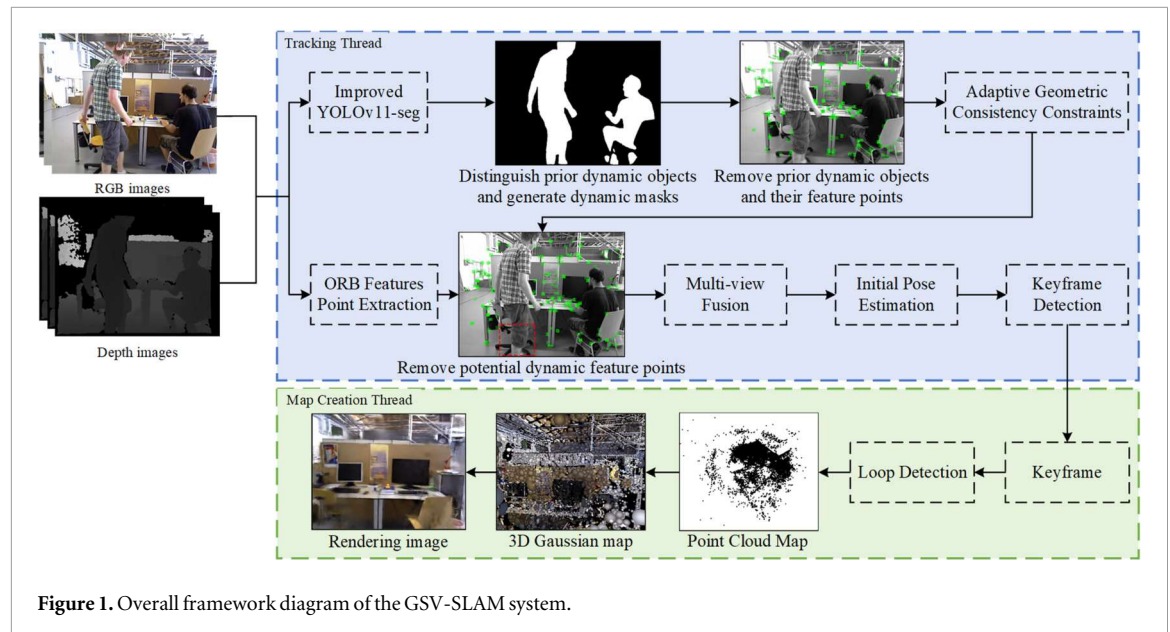


Figure 1. Overall framework diagram of the GSV-SLAM system.

semantically weighted 3D Gaussian field representation that significantly enhances scene understanding and mapping stability. SplatAM [31] employs contour tracking strategies and geometric prior clues to adaptively generate Gaussian point clouds in reconstruction-deficient regions, mitigating the hole-punching issues inherent in traditional reconstruction. However, its tight coupling between tracking and mapping threads limits parallel efficiency and real-time performance. MonoGS [32] proposes a mapping strategy for monocular cameras based on keyframe depth optimization and multi-scale Gaussian fusion. By adaptively updating Gaussian distributions and sparse reconstruction constraints, it achieves high-precision 3D Gaussian field reconstruction with only monocular input. Photo-SLAM [33] decouples ORB-SLAM3's real-time pose estimation from 3DGS mapping, introducing an image pyramid optimization mechanism to enhance Gaussian distribution updates, enabling efficient reconstruction on embedded devices.

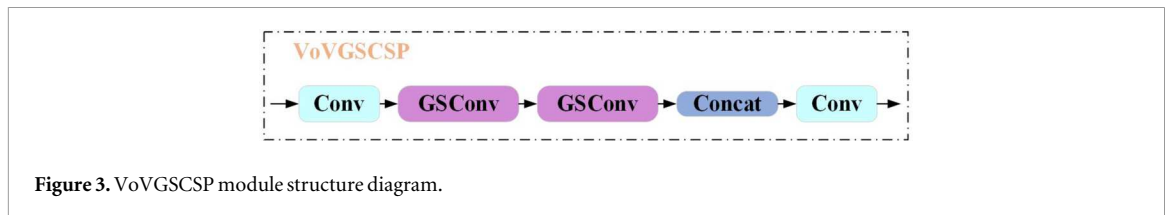
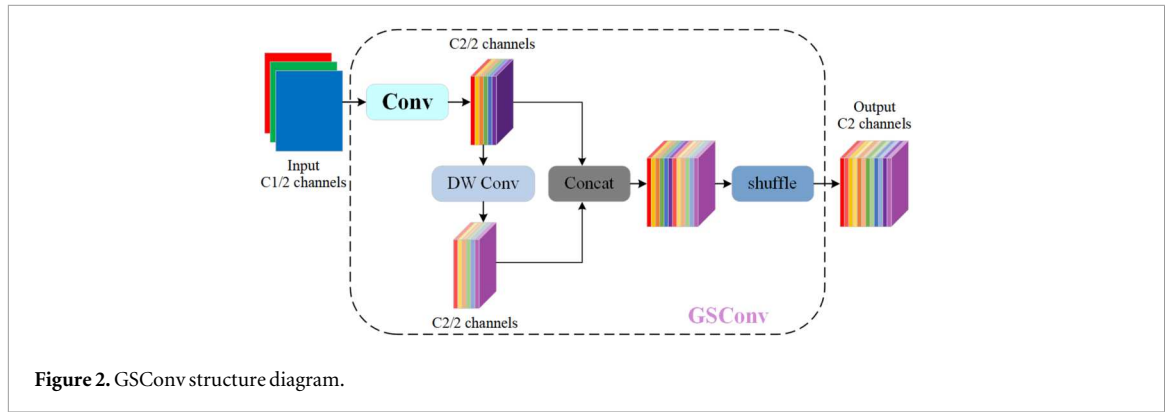
Although the aforementioned methods demonstrate strong performance in static or low-motion environments, they still face robustness challenges in real-world scenarios with complex moving objects. To address this challenge, this paper proposes a dynamic mask generation mechanism based on semantic segmentation and inter-frame spatial geometric consistency. This mechanism identifies and removes features from dynamic regions, significantly enhancing the system's adaptability and stability in dynamic environments.

3. Method

This paper proposes a 3D Gaussian splatting visual SLAM system suitable for dynamic scenes. The overall system architecture is illustrated in figure 1. Leveraging the visual odometry provided by ORB-SLAM3, the method obtains the camera's initial pose. During the front-end tracking process, GSV-SLAM incorporates a dynamic feature rejection strategy. It comprehensively utilizes semantic segmentation information extracted by the YOLOv11 network alongside adaptive geometric consistency constraints to effectively identify and filter out unstable feature points within dynamic regions. Subsequently, the system concurrently performs camera pose estimation and background reconstruction within the current image frame. It selects representative frames as keyframes based on inter-frame viewpoint variations and feature distribution. In the backend mapping module, a set of determined keyframe poses and corresponding point clouds are utilized to construct a high-quality 3D Gaussian map. Through a differentiable rendering mechanism, Gaussian distribution parameters are jointly optimized to achieve high-precision, highly consistent map reconstruction in dynamic environments.

3.1. Improved YOLOv11 for object detection and semantic segmentation

This work introduces a structurally optimized YOLOv11 model. The proposed improved system is structured around four core modules: the input layer, backbone network, neck network, and detection head. Specifically, within the Neck component, we introduce GSConv [34], a novel convolutional operator that partially replaces standard convolution (SC) to strike a better balance between model accuracy and speed. In traditional convolutional neural network (CNN) backbone architectures, spatial information in images is progressively converted into channel information during layer-by-layer processing. However, this process often leads to



semantic information loss due to repeated spatial compression and channel expansion in feature maps. Simultaneously, the high computational complexity of standard convolution significantly reduces inference speed as model parameters increase. To address this, researchers proposed Depthwise Separable Convolution (DSC), which decomposes traditional convolution into two stages: spatial convolution per channel and pointwise convolution. Processing each channel independently substantially reduces computational load while improving overall efficiency. Since the spatial convolution stage operates independently within each channel, cross-channel feature interactions rely on the subsequent pointwise convolution. Consequently, compared to standard convolution, its feature extraction and fusion capabilities are somewhat diminished.

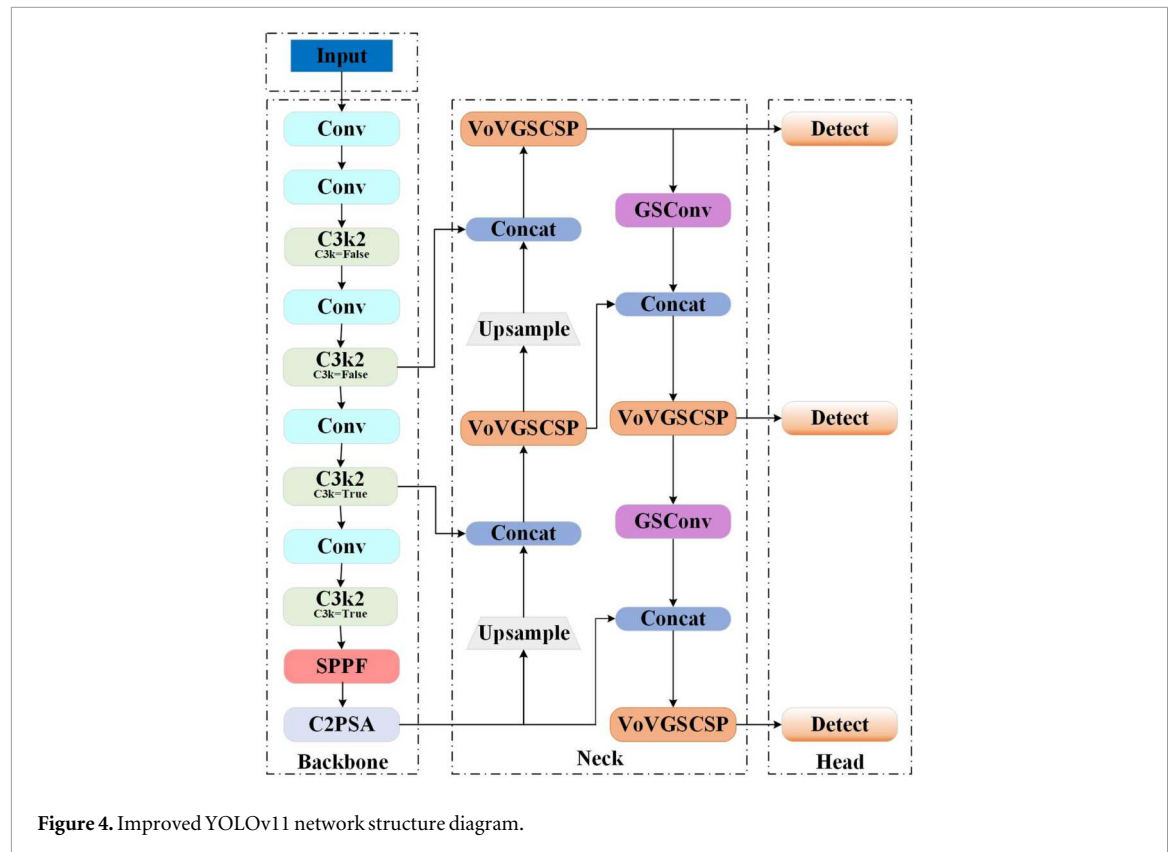
The GSConv convolution method combines the advantages of SC and DSC, preserving inter-channel interactions while maintaining relatively low computational complexity, thus achieving a better balance between accuracy and speed. As shown in figure 2, the GSConv architecture includes the following key steps: The input feature map first passes through a 1×1 standard convolution layer to adjust the number of channels and feature dimensions, with the output channel count being $C_2/2$ of the original input channels. Next, the output features are evenly split into two channel branches. One branch is directly retained as baseline features, while the other is fed into a Depthwise Separable Convolution (DWConv) layer for local spatial feature extraction, enhancing feature expressiveness without significantly increasing computational load. Subsequently, the two branch outputs are combined by channel-wise fusion, enabling the integration of information from the convolutional and depthwise convolutional layers. Finally, a channel shuffle operation rearranges the merged channels to break static grouping, enhance information exchange and feature reuse capabilities, thus improving overall feature representation performance. This further boosts inter-channel information interaction. The process of replacing standard convolutions with GSConv can be formally expressed as:

$$YOLOv11(x) = GSConv(SC(x)) = Shuffle\left(\sum_{i=1}^n Conv(SC(x_i))\right) \quad (1)$$

where, $YOLOv11(x)$ denotes the output of the YOLOv11 model, where x represents the input feature map.

Although GSConv can effectively enhance model efficiency, its extensive application across the entire network structure may lead to increased network depth and obstructed data flow, thus significantly extending the model's inference time. To address this issue, this paper introduces the GSConv structure only at several key stages within the YOLOv11 framework, aiming to balance performance and efficiency. In the neck network, a GSConv-based feature fusion module, VoVGSCSP, is employed to replace the original C3K2 module. By introducing more efficient convolutional operators, the VoVGSCSP module further reduces the computational complexity during feature fusion. As shown in figure 3, the VoVGSCSP structure significantly alleviates the model burden while maintaining comparable performance to traditional modules.

Building on this improvement, we further introduce the SlimNeck module to structurally enhance the neck of YOLOv11. Derived from the Efficient Layer-wise Attention Network (E-ELAN), SlimNeck incorporates a lightweight attention mechanism that adaptively enhances informative feature channels while suppressing



irrelevant background regions. This feature recalibration strategy effectively reduces interference from noisy backgrounds and strengthens the network's ability to focus on salient object structures. As a result, the module greatly enhances the system's stability and positioning precision when dealing with complex visual tasks such as camera pose estimation and dynamic object tracking. The modified YOLOv11 framework is illustrated in figure 4.

To test the model's detection accuracy, this paper constructs a training dataset for dynamic scenes. The dataset combines curated public datasets with independently collected real-world scenarios to ensure both data diversity and scene authenticity. Specifically, it includes: sequences containing human motion and object interactions were selected from the TUM RGB-D [35] dataset; typical scene segments with significant dynamic interference were extracted from the BONN RGB-D Dynamic [36] dataset; and image samples covering multi-category dynamic objects from the COCO [37] dataset were incorporated. To further enhance the model's generalization capability in complex real-world dynamic scenarios, this study independently collected a dataset featuring typical dynamic scenes such as pedestrian movement, moving object interference, and object occlusion.

During the data preprocessing stage, all images are uniformly adjusted to the same resolution and re-labeled according to a unified instance segmentation protocol. The final dataset comprises 1347 training images and 345 test images, exhibiting strong diversity in target scale variations, motion states, and occlusion patterns. This diversity effectively mitigates model overfitting risks and enhances detection robustness. Given that humans are typically the primary and most disruptive dynamic factor in indoor scenes, this study prioritized refined annotations for the 'human' category within the indoor dataset. Figure 5 compares instance segmentation performance between the original YOLOv11n-seg and the improved YOLOv11n-seg model in complex indoor dynamic scenes. Both models accurately detect large-scale objects like pedestrians and office equipment, but the improved model demonstrates superior boundary delineation and instance integrity. For small-scale objects and occlusion-prone items (e.g., handheld objects and desktop accessories), the improved model demonstrates stronger detection stability, effectively reducing missed detections and false positives. Furthermore, even in scenarios with multiple coexisting subjects and strong background interference, the improved model maintains high confidence outputs, reflecting superior environmental adaptability and robustness.

Table 1 presents a performance comparison between the original YOLOv11n model and the improved YOLOv11n model. The data in the table shows that the improved model has approximately 10% fewer parameters than the original version, while the computational cost is significantly reduced by about 36%. In terms of detection accuracy, the improved model's precision increased from 94.1% to 96.3%, an increase of 2.1%, and

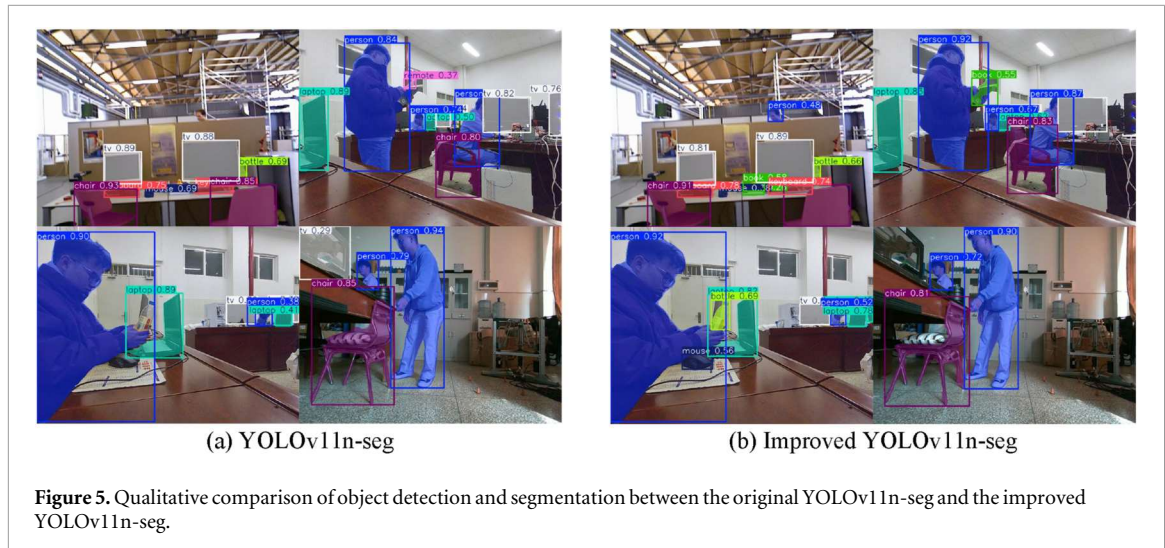


Table 1. Comparison of parameters of different network models.

Model	Parameters/M	Precision/%	mAP(50–95)/%	FLOPs/B
YOLOv11n-seg	2.9	94.1	70.8	9.7
Improved YOLOv11n-seg	2.61	96.3	72.5	6.2

the mAP (50–95) for segmentation tasks also improved from 70.8% to 72.5%, an improvement of 1.7 percentage points. These results demonstrate that by optimizing the feature fusion path and simplifying redundant channel connections, the improved model effectively enhances the representation ability of dynamic targets while maintaining low complexity, achieving a favorable balance between detection accuracy and computational efficiency.

3.2. Adaptive geometric consistency constraints

In real-world environments, the YOLOv11-Seg network may fail to correctly detect or accurately segment certain objects due to occlusions, low visibility, or limitations in semantic category coverage. Furthermore, the segmentation module struggles to identify and exclude feature points belonging to potentially dynamic targets that lack explicit ‘dynamic’ labels in the predefined semantic categories. To tackle the aforementioned challenges, this paper introduces a mechanism combining adaptive geometric consistency constraints to evaluate residual feature points for geometric consistency. The method serves to detect potential dynamic points that were either missed by semantic segmentation or not fully eliminated.

Adaptive geometric consistency constraints are illustrated in figure 6. O_1 and O_2 denote the centers of the two cameras, I_1 represents the reference frame image, and I_2 denotes the current frame image. L_1 and L_2 denote the polar lines associated with the two image frames, e_1 and e_2 denote the poles, P is a point in three-dimensional space, p' and p'' denote the next point positions reached by point P . The corresponding projection points of point P on the two image frames are denoted as P_1 and P_2 , respectively, whose homogeneous coordinates are expressed as follows:

$$P_1 = [x_1, y_1, 1]^T \quad (2)$$

$$P_2 = [x_2, y_2, 1]^T \quad (3)$$

where variables x and y indicate the pixel coordinate values. The polar line L_2 can be calculated using the fundamental matrix F as follows:

$$L_2 = [X, Y, Z]^T = FP_1 = F[x_1, y_1, 1]^T \quad (4)$$

In the equation, X, Y, Z denote line vectors. The fundamental matrix F is estimated using the RANSAC algorithm on matched ORB feature point pairs from the preceding and subsequent frames. Given the known correspondence between P_1 and P_2 in frames I_1 and I_2 , the polar line error between the actual position of the projected point in the current frame and the ideal polar line is:

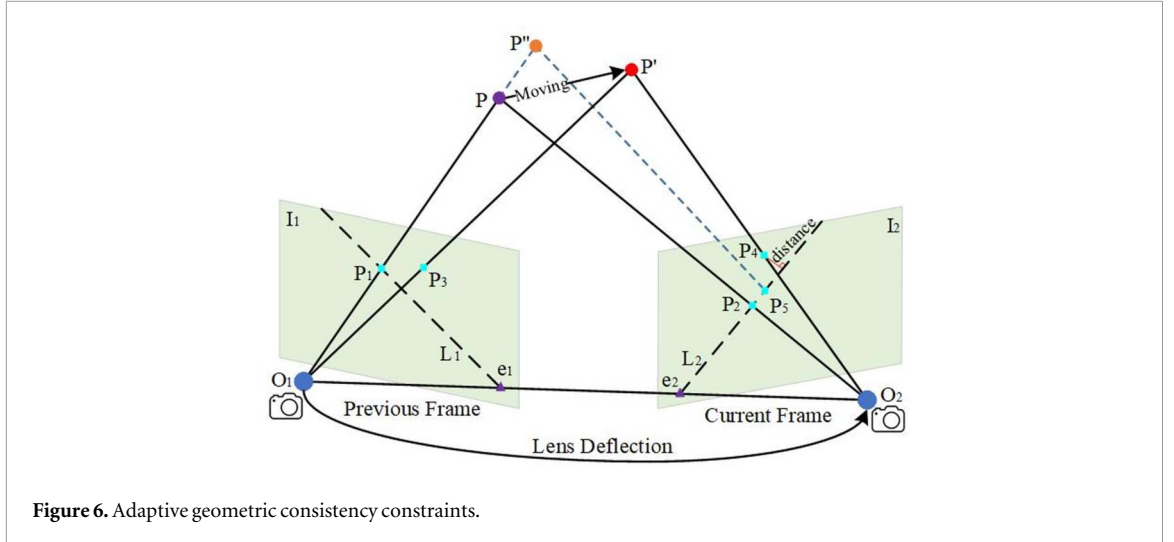


Figure 6. Adaptive geometric consistency constraints.

$$D_1 = \frac{|P_2^T F P_1|}{\sqrt{X^2 + Y^2}} \quad (5)$$

Equation (6) quantifies the pixel-level deviation of a projected point from its ideal geometric constraint. The static or dynamic nature of a point is determined by comparing its polar line error, D_1 , against a predefined threshold θ . Ideally, the reprojection offset for a static point P should be zero, but in practice, noise from the environment and sensors introduces minor displacements. Thus, a point is classified as static provided that $D_1 \leq \theta$. As illustrated in figure 6, when point P moves to P' , the resulting error D_1 between P_4 and the ideal polar line L_2 exceeds θ , correctly identifying it as dynamic. A degenerate case occurs, however, when P moves to a position like P'' along the ray $\overline{O_1P}$. Here, the projection P_5 of P'' onto frame I_2 still remains on L_2 , causing the polar line error to satisfy the static condition and leading to misclassification. Therefore, relying solely on epipolar geometry constraints to identify dynamic feature points has limitations. To overcome this issue, a novel approach is presented which leverages both polar geometry and depth consistency constraints to effectively filter dynamic features. This approach identifies and removes potential dynamic feature points not covered by prior semantic information. By employing a weighted mechanism to calculate a unified motion consistency score for each feature point, it effectively reduces false negatives caused by dynamic feature points moving along the camera's optical center. Let Π denote the camera's projection transformation matrix, which maps 3D spatial points to 2D image coordinates. The relationship between pixels and 3D spatial points is given by:

$$P_2 = \Pi(P) = \left(\frac{x f_x}{z} + c_x, \frac{y f_y}{z} + c_y \right) \quad (6)$$

where, c_x, c_y, f_x, f_y represents the internal parameters of the RGB-D camera, and Π denotes the camera's projection transformation mapping. The projected image coordinates of point P_1 in the current frame are:

$$(u, v) = \Pi(\exp(\hat{\xi}) \Pi^{-1}(P_1)) \quad (7)$$

where, (u, v) denotes the pixel coordinates, Π^{-1} represents the camera's inverse projection transformation mapping, reflecting the process of reconstructing image coordinates into the camera reference frame, and $\exp(\hat{\xi})$ denotes the camera pose transformation from the reference frame to the current frame. The depth error D_2 between the projected point in the reference frame and the current frame is defined as:

$$D_2 = d_1(P_1) - d_2(f_{\text{mean}}(\Pi(\exp(\hat{\xi}) \Pi^{-1}(P_1)))) \quad (8)$$

where, d_1 and d_2 represent the depth mappings corresponding to pixels on reference frame I_1 and current frame I_2 , respectively, while f_{mean} denotes the mean estimation function. To more accurately represent depth variation, this paper employs a depth averaging strategy. A projection point's depth is estimated by averaging values from a 3×3 pixel window centered on its coordinates. To better characterize the impact of depth error and polar line error on motion consistency, this paper introduces an additional weighting parameter to balance their relative importance. The final composite motion score for a feature point is determined by the weighted combination of these two errors:

$$D_s = \lambda D_1 + \omega D_2 \quad (9)$$

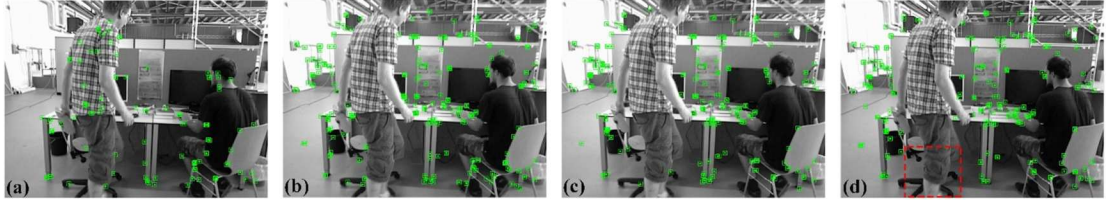


Figure 7. Removal of dynamic feature points. (a) ORB-SLAM3; (b) YOLOv11 semantic segmentation; (c) epipolar geometric constraints; (d) adaptive geometric consistency constraints.

$$\lambda = \exp\left(\frac{D_1}{D_{th1}} - 1\right) \quad (10)$$

$$\omega = \exp\left(\frac{D_2}{D_{th2}} - 1\right) \quad (11)$$

where, λ and ω represent the adaptive weighting factors corresponding to the polar line error D_1 and depth error D_2 , respectively. These factors reflect the relative importance of each error term in the final motion consistency score and dynamically adjust their weight values based on the magnitude of their respective errors. D_{th1} and D_{th2} denote the tolerance thresholds for pole geometry constraint error and depth error, respectively. Based on error statistics and empirical settings, when either the pole geometry constraint error D_1 or depth error D_2 exceeds its corresponding threshold, this indicates the spatial point may be in motion. Consequently, the system assigns higher weights to the relevant error metrics to better evaluate the displacement characteristics of the feature point. Through experimentation and statistical analysis, a threshold for static points, denoted as D_{th} , can be determined. For each matched feature pair P in the current frame, a composite motion score D_s is computed. A point with $D_s > D_{th}$ is deemed dynamic and subsequently discarded. Otherwise, it is retained as a static one.

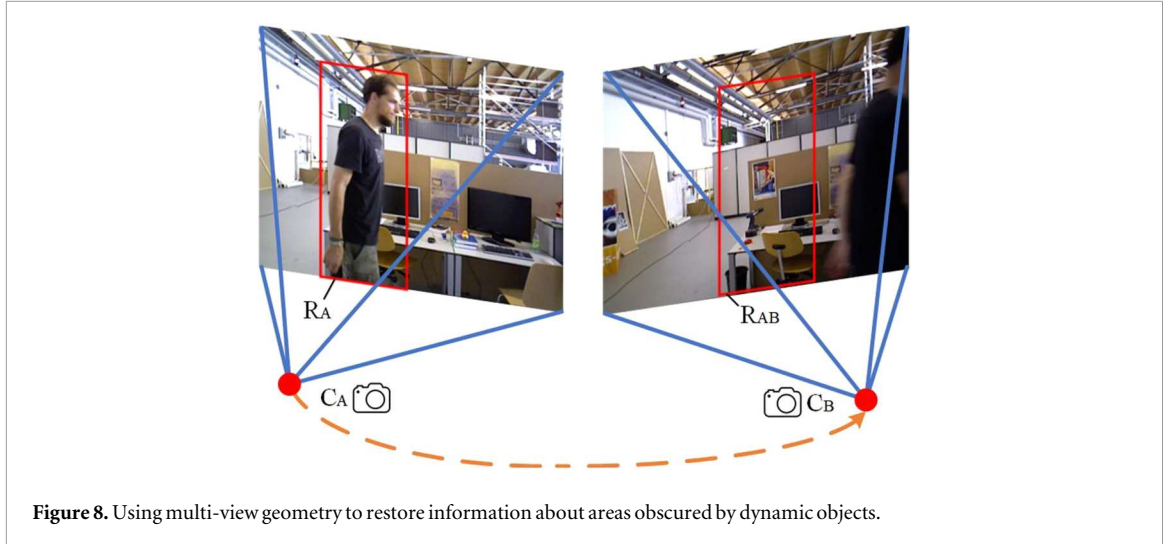
Figure 7 compares the performance of different algorithms in removing dynamic features within the same frame, using the fr3_w_xyz sequence from a challenging high-dynamic dataset. Figure 7(a) depicts the feature point distribution extracted by the original ORB-SLAM3 algorithm. Without incorporating any motion handling mechanism, it extracted numerous feature points on moving human subjects within the scene. This led to inter-frame mismatches that severely compromised the system's localization accuracy and stability. Figure 7(b) shows an improved version based on ORB-SLAM3 with integrated YOLOv11 semantic segmentation. Dynamic feature points on the human body have been effectively removed, but some feature points on potential dynamic objects (e.g., moving chairs) remain undetected. Figure 7(c) demonstrates the combined removal effect of YOLOv11 semantic segmentation and epipolar geometry constraints, some feature points on the moving chair have not been completely removed. Figure 7(d) illustrates the dynamic feature removal capability of the GSV-SLAM algorithm, which combines YOLOv11 semantic segmentation and an adaptive geometric consistency mechanism. This approach further filters out keypoints associated with potentially dynamic regions. The algorithm substantially improves the reliability of the remaining feature points, delivering more stable observations for subsequent frame matching and pose estimation. The system's capabilities in challenging dynamic settings are significantly enhanced, improving both robustness and mapping fidelity.

3.3. Semantic-aware gaussian filter

This study presents a mapping framework that leverages semantic masks of moving objects and multi-view geometric constraints. During feature extraction, semantic cues from YOLOv11-seg are employed to remove keypoints falling within motion regions, simultaneously exclude these dynamic regions during the mapping process.

While removing dynamic targets, we must further address the reconstruction of appearance in regions obscured by them. Dynamic objects frequently obscure the background, creating gaps in feature detection that compromise subsequent mapping integrity. To restore these occluded regions, we employ multi-view geometry techniques. By leveraging visual consistency across different viewpoints, we estimate and reconstruct appearance information for the obscured areas, thereby compensating for feature loss caused by dynamic occlusion.

As shown in figure 8, certain regions in the current view C_A cannot be reconstructed due to occlusion by moving objects, resulting in blank areas. To fill these gaps, information from alternative viewpoints (e.g., C_B) can be utilized. By establishing common views through camera poses, the system enables the reconstruction of occluded region geometry. The recoverable region R_{AB} of the reference viewpoint C_B is projected onto the current viewpoint C_A via the attitude transformation T_{AB} . The recovered region $\tilde{R}_A$ can be expressed as follows:



$$\tilde{R}_A = R_A \cup \bigcup_{B=1}^N T_{AB}(R_{AB}) \quad (12)$$

where, N indicates the count of reference viewpoints, T_{AB} denotes the relative pose between the current viewpoint C_A and viewpoint C_B , R_A indicates the occluded region of dynamic objects in the current viewpoint C_A , and R_{AB} represents the region in viewpoint C_B that can be used to complete the occlusion in viewpoint C_A .

This study introduces a smoothing filter to post-process semantically filtered images, thereby further enhancing mapping quality. The approach first identifies regions containing dynamic objects in each image frame using a semantic segmentation network. These regions are then processed with color smoothing to blend them with the surrounding environment. This operation aims to suppress unstable point clouds that may arise from residual effects of dynamic objects, thus preventing them from interfering with the final mapping results. Despite removing dynamic objects and restoring occluded regions through semantic cues, some portions of the image may continue to produce unstable point cloud observations. The smoothing step mitigates unnecessary Gaussian point formation in these regions, improving global mapping consistency.

The smoothing filter operates under the following formulation: Denote the color of the target region as $C_\alpha(x)$ and that of its adjacent background as $C_\beta(x)$. The filtering process is implemented through the following expression:

$$C_\delta(x) = \eta C_\beta(x) + (1 - \eta) C_\alpha(x) \quad (13)$$

where η denotes the smoothing coefficient, determining the extent of the applied smoothing. In the generation of Gaussian point clouds, regions marked for smoothing are omitted from consideration. Specifically, when the color distribution of a region closely matches that of its surrounding areas, no Gaussian points are produced there. This strategy greatly mitigates the impact of dynamic objects on reconstruction, resulting in a more stable and visually coherent point cloud. The Gaussian point cloud G generated from the smoothed image can be expressed as:

$$G = \{g_i | C_\delta(x_i) \neq C_\delta(x_{\text{dynamic}})\} \quad (14)$$

where, g_i denotes each point in the Gaussian point cloud, x_i represents the pixel position of that point, and $C_\delta(x_{\text{dynamic}})$ indicates the color of the dynamic object region. This design helps mitigate visual artifacts and reduce image blur. Furthermore, these dynamic regions are excluded from Gaussian initialization in the current frame.

Figure 9 demonstrates the rendering quality comparison among SplaTAM [31], MonoGS [32], Photo-SLAM, and GSV-SLAM algorithms when reconstructing Gaussian maps on two highly dynamic dataset sequences: fr3_walking_xyz and fr3_walking_halfsphere. The figures reveal that maps generated by SplaTAM, MonoGS, and Photo-SLAM exhibit noticeable motion artifacts, resulting in blurred maps and inaccurate structural representation. In contrast, the proposed GSV-SLAM algorithm effectively mitigates these issues. It produces maps with clear structures and sharp boundaries, successfully eliminating interference from dynamic objects to construct static scene maps. The findings clearly confirm that the designed dynamic feature rejection mechanism is effective when applied to real-world dynamic scenarios.

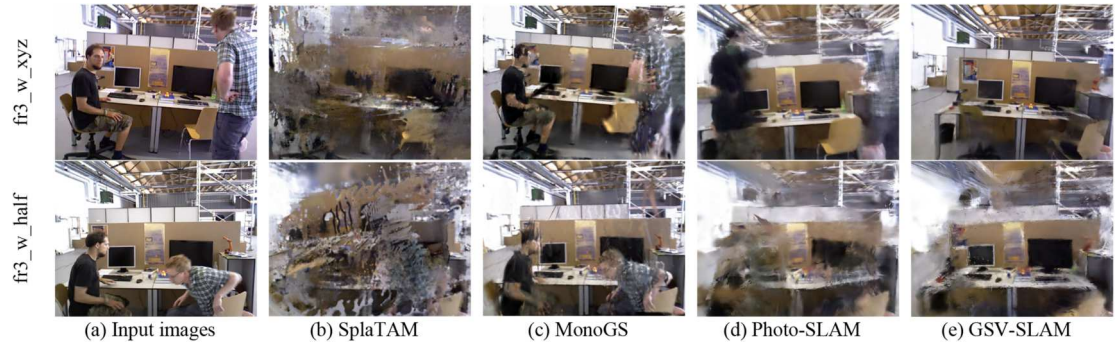


Figure 9. Dynamic scene visualization comparison.

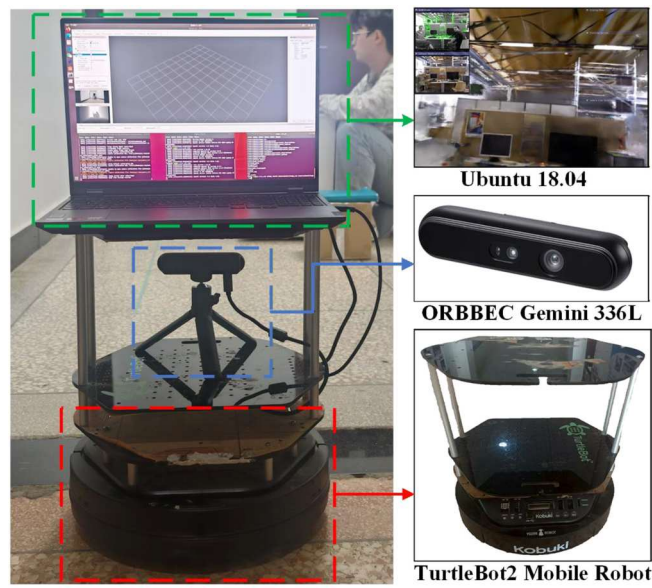


Figure 10. Experimental platform.

4. Experiment

To validate the effectiveness of the proposed algorithm, this study employs a Turtlebot2 mobile robot based on the Robot Operating System (ROS) as the carrier platform. The host machine is a laptop that integrates an AMD Ryzen 7 5800H processor and an NVIDIA RTX 3060 GPU with 6 GB of video memory, supported by 16 GB of RAM and Ubuntu 18.04. The visual sensor employed is the ORBBEC Gemini 336L full-scene stereo structured light 3D camera, capturing synchronized color and depth images for visual SLAM processing. For datasets, this paper employs the TUM RGB-D and BONN RGB-D dynamic datasets, widely used in visual SLAM evaluation. These datasets contain well-aligned color and depth images, featuring rich dynamic scenes and high-precision trajectory annotations. They effectively validate the system's robustness and mapping accuracy in static, low-dynamic, and high-dynamic environments. The experimental platform is shown in figure 10.

4.1. Comparison of experimental results from public datasets

To quantitatively evaluate the performance of the proposed system in dynamic scenarios, comparative experiments were conducted. The assessment constituted Absolute Trajectory Error (ATE) and Relative Pose Error (RPE) as the principal criteria for performance evaluation. A comparison with the classical ORB-SLAM3 framework validated the superior robustness of our method in handling dynamic scenes. The ATE measures the global deviation between the entire estimated trajectory and the ground truth, reflecting the system's overall positioning accuracy and trajectory preservation capability. The RPE computes the relative pose discrepancy

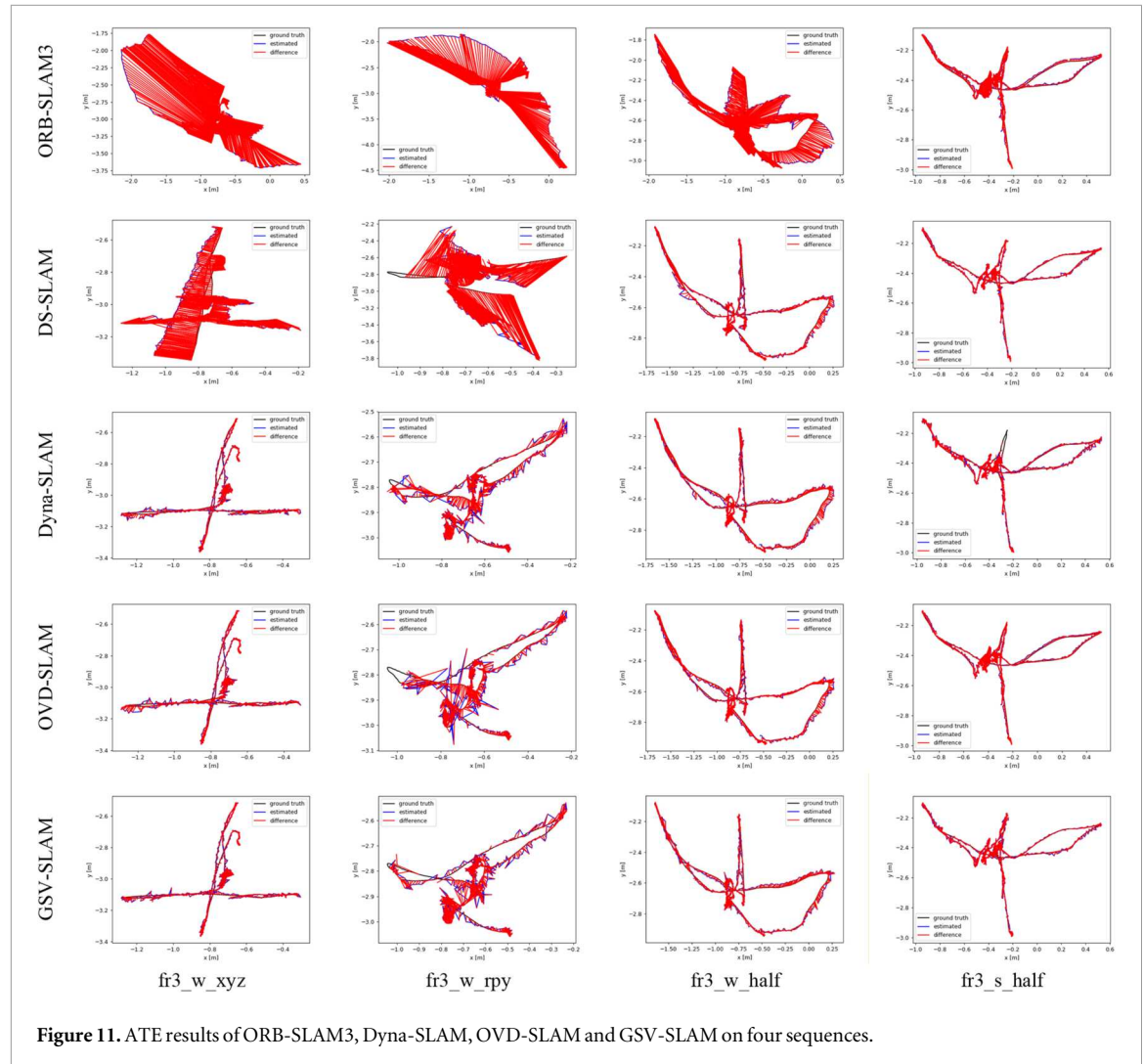


Figure 11. ATE results of ORB-SLAM3, Dyna-SLAM, OVD-SLAM and GSV-SLAM on four sequences.

between pairs of poses over fixed time intervals, primarily serving to analyze the system's local tracking accuracy.

Figure 11 and figure 12 show the absolute trajectory error plots and relative trajectory error plots for ORB-SLAM3 [3], DS-SLAM [38], Dyna-SLAM [19], OVD-SLAM [22], and GSV-SLAM. In highly dynamic scenes, the camera trajectory estimated by ORB-SLAM3 exhibits significant deviation from the true trajectory, with noticeable positioning drift occurring in certain areas. In contrast, DS-SLAM, DynaSLAM, and OVD-SLAM demonstrate relatively robust performance in highly dynamic environments. However, their dynamic feature point filtering mechanisms remain incomplete, resulting in residual interference.

As shown in figure 12, the black, blue, and red curves represent the camera trajectory, the estimated trajectory, and the error between the actual trajectory and the estimated trajectory, respectively. The figure demonstrates that DS-SLAM achieves significant improvements over ORB-SLAM3, though errors remain relatively large in the fr3_w_xyz and fr3_w_rpy sequences. Dyna-SLAM demonstrates robust performance across all datasets, reducing trajectory errors consistently. However, it suffers from severe frame loss on the fr3_w_rpy and fr3_s_half sequences, resulting in the discard of numerous camera poses. Although OVD-SLAM shares the frame drop drawback with Dyna-SLAM on fr3_w_rpy, it significantly outperforms Dyna-SLAM on the static dataset fr3_s_half. In contrast, the proposed GSV-SLAM achieves more accurate alignment of real trajectories, demonstrating higher overall consistency and precision. The proposed method demonstrates superior localization performance in highly dynamic scenarios.

To further validate the effectiveness of the proposed algorithm, this paper quantitatively compares the absolute trajectory error (ATE) of GSV-SLAM with mainstream dynamic SLAM methods such as ORB-SLAM3 [3], DS-SLAM [37], Dyna-SLAM [19], OVD-SLAM [22], Blitz-SLAM [39], and YVG-SLAM [40]. Key evaluation metrics include root mean square error (RMSE), median, and standard deviation (SD). Algorithms marked with an asterisk (*) do not provide open-source code; thus, this paper uses data published in the

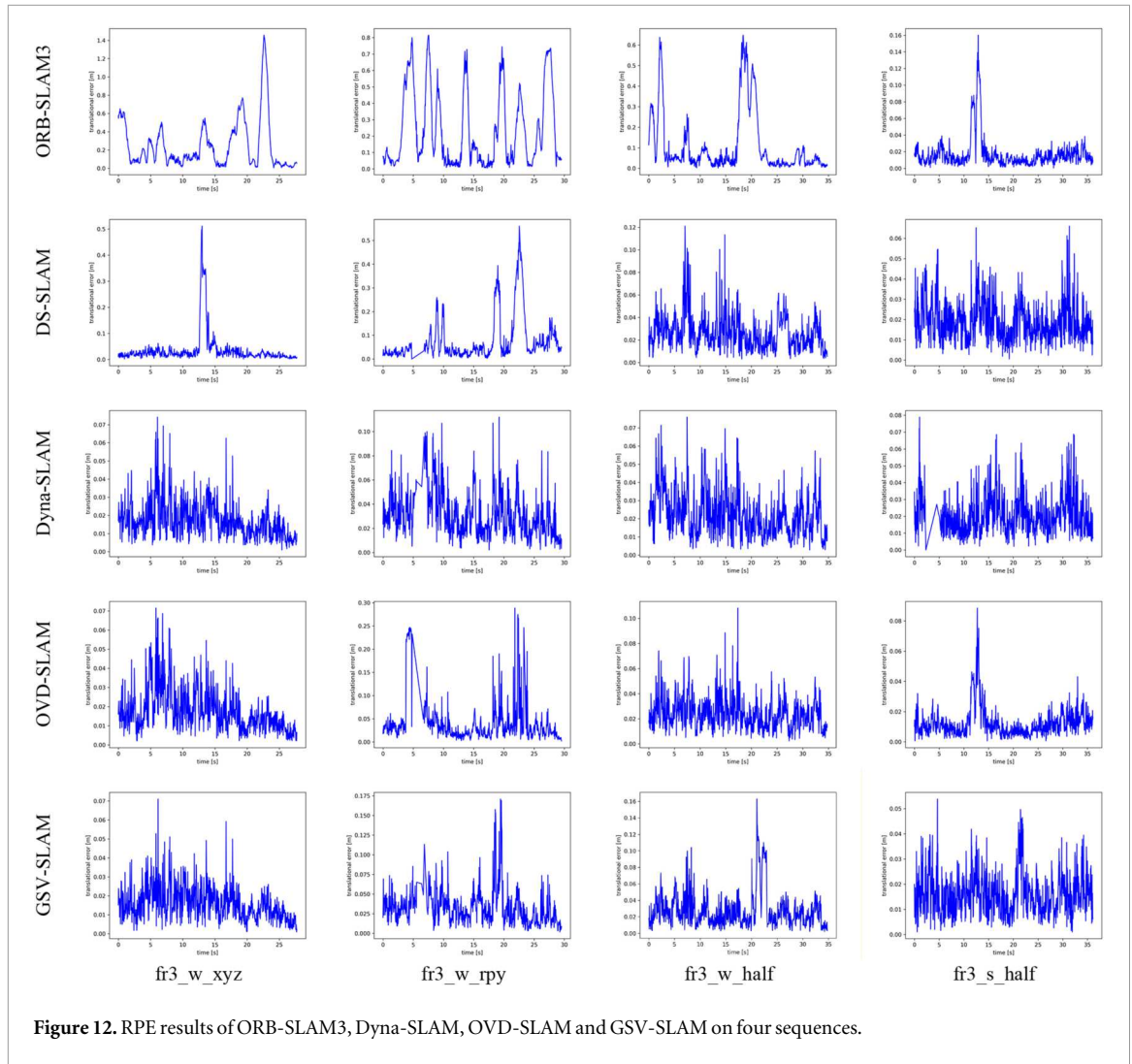


Figure 12. RPE results of ORB-SLAM3, Dyna-SLAM, OVD-SLAM and GSV-SLAM on four sequences.

Table 2. ATE (m) Comparison Between ORB-SLAM3 and GSV-SLAM.

Sequence	ORB-SLAM3			GSV-SLAM			Improvements (%)		
	RMSE	Median	STD	RMSE	Median	STD	RMSE	Median	STD
fr2_desk	0.0202	0.0186	0.0075	0.0198	0.0168	0.0071	1.98	9.68	5.33
fr2_desk_p	0.3512	0.2923	0.1151	0.0083	0.0068	0.0040	97.64	97.67	96.52
fr3_s_xyz	0.0093	0.0075	0.0045	0.0099	0.0078	0.0051	−6.45	−4.00	−13.33
fr3_s_static	0.0109	0.0094	0.0047	0.0063	0.0047	0.0031	42.20	50.00	34.04
fr3_s_rpy	0.0222	0.0119	0.0145	0.0202	0.0115	0.0129	9.01	3.36	11.03
fr3_s_half	0.0271	0.0198	0.0147	0.0158	0.0124	0.0078	41.70	37.37	46.94
fr3_w_half	0.3228	0.2801	0.1168	0.0210	0.0154	0.0108	93.49	94.50	90.75
fr3_w_rpy	0.7464	0.5046	0.4111	0.0283	0.0195	0.0167	96.21	96.14	95.94
fr3_w_static	0.0731	0.0173	0.0644	0.0059	0.0055	0.0031	91.93	68.21	95.19
fr3_w_xyz	0.7008	0.4648	0.4352	0.0131	0.0091	0.0063	98.13	98.04	98.55

original papers for comparison. Tables 2 and 3 present the absolute and relative trajectory error evaluation results of GSV-SLAM versus ORB-SLAM3 on the TUM RGB-D dataset.

GSV-SLAM achieves an average RMSE reduction of 95.48% compared to ORB-SLAM3 in high-dynamic scenes and 17.69% in low-dynamic datasets. This is because ORB-SLAM3 is primarily designed for static scenes, where it demonstrates strong localization and mapping performance. Table 4 systematically compares the performance of GSV-SLAM against multiple advanced dynamic SLAM methods, with optimal results highlighted in bold and second-best results underlined. Experimental results demonstrate that our method achieves higher trajectory estimation accuracy and stronger system stability in most dynamic scenarios, with overall

Table 3. RPE (m) Comparison Between ORB-SLAM3 and GSV-SLAM.

Sequence	ORB-SLAM3			GSV-SLAM			Improvements (%)		
	RMSE	Median	STD	RMSE	Median	STD	RMSE	Median	STD
fr2_desk	0.0082	0.0069	0.0038	0.0079	0.0068	0.0035	3.66	1.45	7.89
fr2_desk_p	0.0932	0.0616	0.0567	0.0081	0.0056	0.0047	91.31	90.91	91.71
fr3_s_xyz	0.0115	0.0088	0.0058	0.0125	0.0094	0.0067	−8.7	6.82	−15.52
fr3_s_static	0.0104	0.0080	0.0049	0.0076	0.0056	0.0039	26.92	30.00	20.41
fr3_s_rpy	0.0274	0.0169	0.0164	0.0253	0.0164	0.0149	7.66	2.96	9.15
fr3_s_half	0.0263	0.0175	0.0196	0.0194	0.0150	0.0096	26.24	14.29	51.02
fr3_w_half	0.1851	0.0459	0.1487	0.0247	0.0186	0.0128	86.66	59.48	91.39
fr3_w_rpy	0.3323	0.1239	0.2333	0.0437	0.0284	0.0269	86.85	77.08	88.47
fr3_w_static	0.1044	0.0144	0.0936	0.0105	0.0083	0.0052	89.94	42.36	94.44
fr3_w_xyz	0.3653	0.1329	0.2693	0.0170	0.0144	0.0093	95.35	89.16	96.55

performance matching or surpassing existing mainstream methods. This fully reflects the competitive advantage of the proposed algorithm in dynamic environments.

To evaluate the positioning performance of GSV-SLAM in highly dynamic scenes, quantitative experiments were conducted on the BONN RGB-D Dynamic dataset. This dataset contains numerous fast-moving targets and occlusion interference, posing significant challenges to the system's dynamic processing capabilities. Four representative sequences, namely *bonn_moving_nonobstructing_box*, *bonn_crowd*, *bonn_balloon*, and *bonn_kidnapping_box*, were selected for evaluation. Using absolute trajectory error as the metric, the proposed method was compared with other SLAM approaches including ORB-SLAM3, DS-SLAM, DynaSLAM, YVG-SLAM and OVD-SLAM. Results are summarized in table 5. Overall, in scenarios with significant dynamic interference, most methods exhibit increased positioning errors, while GSV-SLAM delivers relatively stable results across all sequences.

4.2. Ablation studies

To validate the actual contribution of each dynamic feature filtering module to system performance, ablation studies were conducted on multiple TUM dynamic sequences. The experiments evaluated three configurations: GSV-SLAM(Y) employing only the improved YOLOv11n-seg for dynamic object detection; GSV-SLAM(Y+G) combining YOLO with epipolar geometry constraints; and GSV-SLAM(Y+G+D) further integrating a geometric consistency detection strategy. Table 6 presents absolute trajectory error (ATE) results for different configurations, using RMSE and standard deviation (SD) as evaluation metrics, with optimal results highlighted in bold.

The table reveals a consistent improvement in positioning accuracy as dynamic constraint strategies are progressively introduced. Compared to GSV-SLAM(Y) using only semantic detection, integrating pair-wise geometric constraints significantly reduces system error across all sequences. This demonstrates that geometric constraints eliminate feature points from potentially dynamic objects, enhancing pose estimation stability. Further incorporating geometric consistency detection enables GSV-SLAM(Y+G+D) to achieve the best performance in most sequences. For instance, in the *fr3_w_rpy* sequence, RMSE decreased from 0.0721 to 0.0283, representing an error reduction exceeding 60%. In the *fr3_w_half* sequence, where dynamic interference is more pronounced, system errors also decreased significantly. This demonstrates that the geometric consistency strategy can further identify potential dynamic regions, reducing the interference of moving objects on trajectory optimization.

This paper statistically analyzes the segmentation and tracking runtime of various dynamic SLAM methods. The experimental results are shown in table 7. All experiments were conducted on the same hardware platform to ensure fair comparison.

As shown in table 7, the traditional ORB-SLAM3 achieves an average tracking time of only 25.4 ms due to the absence of a semantic segmentation module, demonstrating high operational efficiency. However, it is susceptible to interference from moving objects in dynamic scenes, thereby compromising positioning accuracy and mapping quality. After incorporating a deep learning segmentation network, the system's computational overhead increases significantly. Specifically, DynaSLAM employs Mask R-CNN for dynamic object detection, with single-frame segmentation taking 196.3 ms, resulting in an overall tracking time of 563.8 ms—unsuitable for real-time applications. In contrast, the proposed GSV-SLAM leverages an optimized lightweight YOLOv11n-seg network. It maintains dynamic object detection capabilities while effectively controlling computational overhead, achieving a single-frame segmentation time of 37.6 ms and an average tracking time of 67.7 ms. Although its average tracking time exceeds that of ORB-SLAM3, its accuracy significantly outperforms

Table 4. ATE (m) comparison between GSV-SLAM and state-of-the-art dynamic SLAM methods.

Sequence	DS-SLAM		Dyna-SLAM		Blitz-SLAM*		OVD-SLAM		YVG-SLAM*		GSV-SLAM	
	RMSE	STD	RMSE	STD	RMSE	STD	RMSE	STD	RMSE	STD	RMSE	STD
fr3_s_xyz	0.0124	0.0112	0.0134	0.0203	0.0148	<u>0.0069</u>	<u>0.0117</u>	0.0102	0.0118	/	0.0099	0.0051
fr3_s_half	0.0137	0.0132	0.0172	0.0092	0.0160	0.0076	0.0168	0.0083	<u>0.0158</u>	0.0075	<u>0.0158</u>	<u>0.0078</u>
fr3_w_half	0.0251	0.0159	0.0268	0.0137	0.0256	0.0126	<u>0.0242</u>	<u>0.0113</u>	0.0285	0.0151	0.0210	0.0108
fr3_w_rpy	0.3107	0.1698	<u>0.0285</u>	0.0158	0.0356	0.0220	0.0349	0.0257	0.0385	0.0242	0.0283	<u>0.0167</u>
fr3_w_static	<u>0.0065</u>	0.0038	0.0074	0.0030	0.0102	0.0052	0.0071	0.0032	/	/	0.0059	<u>0.0031</u>
fr3_w_xyz	0.0247	0.0504	0.0152	0.0075	0.0153	0.0078	<u>0.0147</u>	<u>0.0071</u>	0.0189	0.0091	0.0131	0.0063

Table 5. ATE (m) comparison between GSV-SLAM and state-of-the-art dynamic SLAM methods on the Bonn dataset.

Sequence	ORB-SLAM3		DS-SLAM		Dyna-SLAM		YVG-SLAM*		OVD-SLAM		GSV-SLAM	
	RMSE	STD	RMSE	STD	RMSE	STD	RMSE	STD	RMSE	STD	RMSE	STD
crowd	0.5458	0.4783	0.3524	0.2527	0.0486	0.0336	<u>0.0282</u>	<u>0.0157</u>	0.0293	0.0195	0.0245	0.0125
balloon	0.0729	0.0653	0.0319	0.0145	<u>0.0307</u>	<u>0.0119</u>	0.0303	0.0127	0.0367	0.0191	0.0281	0.0113
mov_non_box	0.3195	0.0951	0.0918	0.0722	0.0518	0.0239	<u>0.0299</u>	0.0198	0.0343	0.0249	0.0231	<u>0.0223</u>
kidnap_box	0.0274	0.0134	0.0305	0.0155	0.0292	0.0159	<u>0.0256</u>	<u>0.0135</u>	0.0268	0.0141	0.0252	0.0126

Table 6. ATE (m) ablation study of different module combinations in GSV-SLAM on the TUM dataset.

Sequence	GSV-SLAM(Y)		GSV-SLAM(Y+G)		GSV-SLAM(Y+G+D)	
	RMSE	STD	RMSE	STD	RMSE	STD
fr3_s_xyz	0.0103	0.0097	0.0101	0.0089	0.0099	0.0051
fr3_s_static	0.0106	0.0096	0.0087	0.0062	0.0063	0.0031
fr3_s_rpy	0.0219	0.0136	0.0208	0.0124	0.0202	0.0129
fr3_s_half	0.0252	0.0163	0.0194	0.0103	0.0158	0.0078
fr3_w_half	0.0365	0.0218	0.0283	0.0138	0.0210	0.0108
fr3_w_rpy	0.0721	0.0622	0.0364	0.0206	0.0283	0.0167
fr3_w_static	0.0086	0.0053	0.0065	0.0038	0.0059	0.0031
fr3_w_xyz	0.0203	0.0131	0.0146	0.0071	0.0131	0.0063

Table 7. Time comparison among different algorithms.

Method	Semantic	Segmentation time (ms)	Average tracking time (ms)
ORB-SLAM3	-	-	25.4
DynaSLAM	Mask R-CNN	196.3	563.8
DS-SLAM	SegNet	43.5	77.6
OVD-SLAM	YOLOv5+Optical Flow	39.2	69.4
GSV-SLAM	Improved YOLOv11n-seg	37.6	67.7

ORB-SLAM3. In summary, GSV-SLAM maintains high operational efficiency while incorporating semantic awareness and geometric constraints, achieving an effective balance between accuracy and real-time performance in dynamic environments.

4.3. Real-world environment experimental evaluation

To validate the effectiveness of the proposed adaptive geometric consistency constraint in real dynamic environments, this paper conducted a visual analysis of the iterative elimination process of dynamic feature points, with the results shown in figure 13.

Figure 13 sequentially displays the ORB-SLAM3 feature point distribution, the dynamic feature point removal results based on improved YOLOv11n-seg instance segmentation, and the feature point filtering effect after introducing geometric consistency constraints. Column (c) of figure 13 demonstrates that the semantic segmentation method effectively removes feature points from most prominent dynamic regions, though feature points on potential dynamic objects remain unremoved. Column (d) shows that introducing adaptive geometric consistency constraints enables the system to effectively identify and eliminate feature points on potential dynamic objects, thereby significantly reducing misassociations caused by dynamic interference.

To validate the positioning performance of the proposed algorithm in real-world dynamic environments, pedestrians walking in a laboratory setting were selected to simulate moving objects in actual environments. The detection effectiveness of this algorithm was evaluated when dynamic objects appeared in the experimental scenario, and the ability to exclude feature points from moving objects was assessed. Rectangular-shaped routes were chosen to simulate rectangular paths in real environments, testing the algorithm's accuracy and robustness during movement along these routes.

Figure 14(a) shows the actual experimental environment and the ground-truth trajectory. Figure 14(b) compares the trajectories estimated by the proposed method against ORB-SLAM3, DS-SLAM, Dyna-SLAM, and OVD-SLAM along the same route. The presence of numerous dynamic feature points, resulting from pedestrian movement during the experiment, posed a significant challenge to all systems. The ORB-SLAM3 algorithm exhibited significant trajectory drift and failed to form a closed loop. Other algorithms also experienced varying degrees of trajectory drift or were unable to complete closed loops. The proposed algorithm eliminates interference from moving objects by filtering out pedestrian features and potential moving object features. This strategy enables complete and precise closed-loop tracking with minimal error, showcasing enhanced positioning reliability and resilience relative to existing techniques operating amid scene dynamics.

Figure 15 visualizes a direct performance comparison between GSV-SLAM and ORB-SLAM3 in filtering pedestrian-borne dynamic features, demonstrating enhanced efficacy in real-world scenarios. The figure reveals that GSV-SLAM accurately identifies and completely removes feature points from dynamic targets such as pedestrians, whereas ORB-SLAM3 exhibits incomplete removal of dynamic feature points. This result

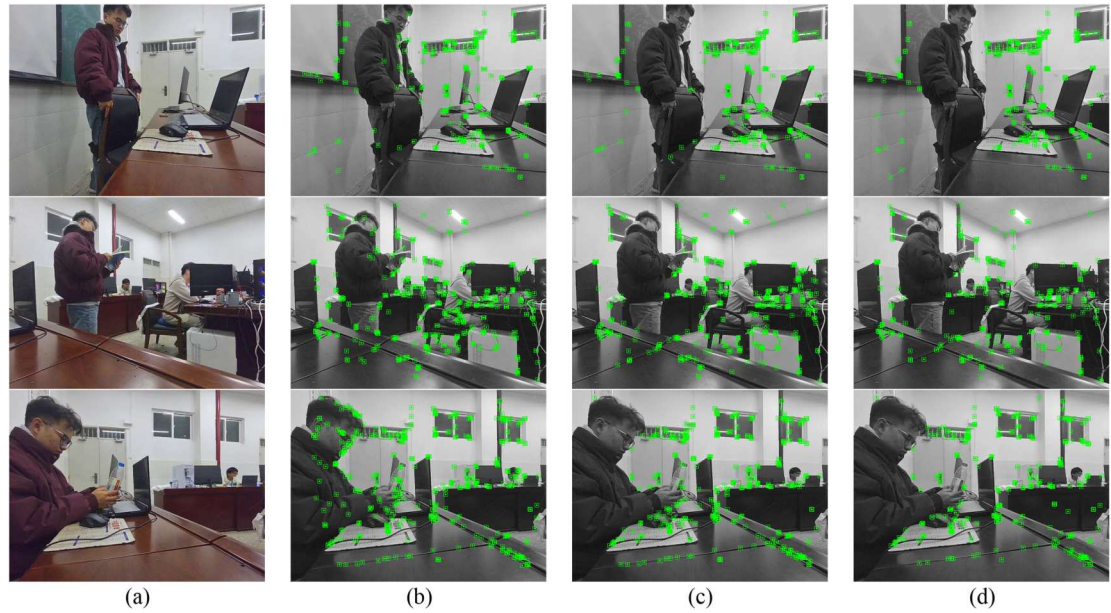
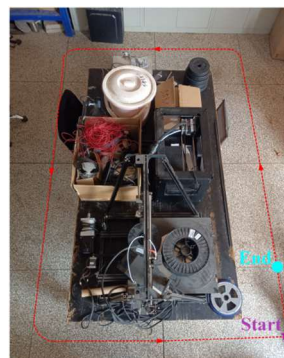
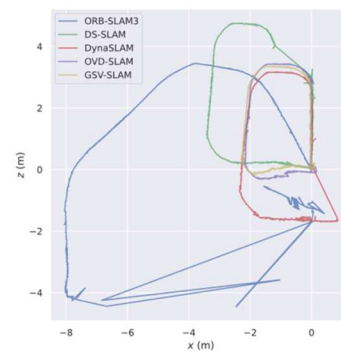


Figure 13. (a) Ground truth image, (b) Feature point distribution using the ORB-SLAM3 algorithm, (c) Feature point distribution after removing dynamic points using YOLOv11n-seg, (d) Feature point distribution after removing dynamic points using YOLOv11n-seg with geometric consistency.



(a) Real trajectory



(b) Trajectory Comparison

Figure 14. Comparison of trajectories generated by different algorithms in real dynamic scenarios.

validates GSV-SLAM's superior performance in real-world dynamic scenarios, demonstrating its ability to effectively suppress dynamic interference, leading to more reliable and precise state estimation.

4.4. 3D Gaussian map construction

Figure 16 presents the 3D Gaussian map reconstruction results of SplatAM [31], MonoGS [32], Photo-SLAM [33], and GSV-SLAM in real dynamic scenes. The experiment selected dynamic scenes featuring multi-person interactions and local occlusions to evaluate the mapping stability and robustness of each algorithm in complex dynamic environments.

The figures reveal that SplatAM and MonoGS effectively reconstruct scene geometry in static regions but exhibit point cloud blurring and drift in areas with significant moving objects. Photo-SLAM's generated map contains numerous dynamic artifacts, where human artifacts intermingle with background point clouds, causing ghosting and voids in local structures that compromise overall map continuity and accuracy. In contrast, GSV-SLAM incorporates semantic constraints and a dynamic feature separation mechanism during mapping. This enables effective identification and removal of Gaussian-distributed points from dynamic objects, thereby preserving the stable static components of the scene. The resulting Gaussian map exhibits continuous structure with clear boundaries, maintaining consistency and integrity even during multi-view synthesis.

Evaluation confirms that GSV-SLAM achieves high-precision reconstruction in static areas, coupled with exceptional robustness and generalization capabilities in dynamic, non-fixed-view scenarios. This method



Figure 15. Dynamic feature point removal in real-world scenarios.



Figure 16. Comparison of rendering results of real datasets.

enables high-quality 3D reconstruction of complex real-world environments, providing a reliable environmental perception foundation for subsequent tasks such as localization, navigation, and semantic understanding.

5. Conclusion

This paper proposes a dynamic SLAM system based on ORB-SLAM3. The system utilizes prior semantic information for target detection to assist the SLAM system in eliminating interference caused by moving objects. For the object detection thread, we propose a lightweight, improved YOLOv11-seg semantic segmentation network that provides the semantic information required in dynamic environments while enhancing detection speed. Additionally, to enhance robustness, the dynamic feature filtering thread incorporates a dual geometric verification mechanism using polarity and depth consistency to identify and suppress dynamic feature points from moving entities. This method maximizes the retention of static feature points in the detection frame while removing all dynamic feature points. Through validation on publicly available RGB-D datasets and real-world scenarios, GSV-SLAM demonstrates significant superiority over the original ORB-SLAM3 in highly dynamic environments while also showing performance improvements in low-dynamic settings. The system further integrates 3D Gaussian splatting technology to achieve high-quality, photorealistic rendering of dynamic scenes. By processing input images with dynamic masks, the system precisely distinguishes static and dynamic regions, performing dense 3D reconstruction solely on static areas to ensure cleaner, more complete scene models. However, limitations persist in large, boundary-free outdoor

environments: 3D Gaussian distributions become sparse in distant regions, and residual artifacts may appear near dynamic areas or occlusion boundaries during rendering. Future research will focus on complex multiscale optimization to enhance mapping accuracy and real-time performance in intricate environments.

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Conflict of interest

The authors declare no competing interests.

Data availability statement

The data obtained in real dynamic environments involve identifiable individuals and indoor scenes, and therefore cannot be shared publicly due to privacy restrictions. The data that support the findings of this study are available upon reasonable request from the authors.

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